and the initial condition that $y = 10$ when $x = 2$.

From the given equation

$$\frac{dy}{dx} = 2x$$

$$y = \int 2x\, dx$$

$$y = x^2 + c$$

We substitute $y = 10$ when $x = 2$, in the above equation

$$10 = 2^2 + c \;\Rightarrow\; c = 6$$

When this value $c$ =6 is substituted we obtain

$$y = x^2 + 6$$

which gives the particular antiderivative desired.

## 11.3 Basic Rules of Integration

### Standard results:

Since integration is the reverse process of differentiation, the basic integration formulae given below can be derived directly from their corresponding derivative formulae from earlier chapter.

| Derivatives | Antiderivatives |
|---|---|
| $\frac{d}{dx}(c) = 0$, where $c$ is a constant | $\int 0\, dx = c$, where $c$ is a constant |
| $\frac{d}{dx}(kx) = k$, where $k$ is a constant | $\int k\, dx = kx + c$ where $c$ is a constant |
| $\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$ | $\int x^n dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$ (Power rule) |
| $\frac{d}{dx}(\log x) = \left(\frac{1}{x}\right)$ | $\int \frac{1}{x}\, dx = \log\lvert x\rvert + c$ |
| $\frac{d}{dx}(-\cos x) = \sin x$ | $\int \sin x\, dx = -\cos x + c$ |
| $\frac{d}{dx}(\sin x) = \cos x$ | $\int \cos x\, dx = \sin x + c$ |
| $\frac{d}{dx}(\tan x) = \sec^2 x$ | $\int \sec^2 x\, dx = \tan x + c$ |



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